Rui Chen

Research Statement

M.Sc. in Pharmacy | Three years of industrial medicinal chemistry | Applying for PhD entry, 2027

1. Background and Direction

I am a medicinal chemist trained in synthesis and structure-based design, with three years of industrial experience in antiviral drug discovery. At Jiangsu Aidea Pharmaceutical I designed and synthesised analogues of a clinical-stage HIV-1 integrase inhibitor, contributed to a long-acting injectable prodrug platform, and took a chiral key intermediate from a laboratory route to a scaleable, lower-cost process. Alongside the bench work I maintained a computational workflow spanning docking, molecular dynamics and ADMET prediction, used to decide which analogues were worth making rather than to replace the decision. That combination — synthesis, computation, and the hard constraints of developability — is what I bring to a doctoral programme.

2. Research Directions I Am Prepared For

I am describing a range of directions rather than a single project: my industrial training was broad, and I would rather state honestly what I can contribute.

(1) Induced proximity and targeted protein degradation. Degraders are, structurally, a conjugation problem: a target-binding ligand, an E3-recruiting element and a linker must be assembled without destroying the binding of either end. My work constructing high-molecular-weight prodrugs (MW > 1500) and my grounding in conjugate chemistry map directly onto this. What I need to develop is the mechanistic side — ternary-complex geometry, cooperativity, hook effects, and how linker length and exit vector govern selectivity. I am particularly interested in E3 ligases beyond CRBN and VHL, where the rules are still being written.

(2) Antiviral medicinal chemistry and developability. Three years on HIV programmes taught me that a molecule is only as good as the path it can survive. I have worked on metabolic-stability and half-life optimisation, solubility rescue, crystal-form identification and process development — the problems that decide whether an active compound reaches a patient. Antiviral chemistry was my employer's need rather than my own choice; it became my subject because whichever problem I am given, I work it properly.

(3) Computational design and ADMET judgement. I use CADD as a decision aid, not a substitute for chemical judgement — choosing which twenty analogues to make out of two hundred proposed, and knowing when a model's prediction should be overruled by assay data.

3. What I Want from a PhD

I chose to return to research because I can now see precisely where my knowledge stops. In industry I learned to execute a programme; what I did not learn is how to explain a molecule's activity at the level of mechanism rather than observation.

From executor to owner of a project. In a company, work arrives as an assigned target and success means meeting it. A doctoral project is different: no one will define my question for me, and no one will tell me when the work is finished. Three years of cross-departmental collaboration taught me that most problems I could not solve alone, and that asking the right person early is not a weakness. Leading a four-person team to a Bronze Award in the AI-for-Science track of a competition hosted by Peking University was the first time I had to define the work as well as do it. I expect to run my own project as a small enterprise — communicating and collaborating rather than simply executing.

Transferable method, and the judgement above it. I want two things from a doctorate that industrial work cannot give me. The first is transferable method — how to reason about molecular design rather than how to execute a given protocol. Designing an induced-proximity molecule, choosing a conjugation strategy, analysing SAR rather than accumulating it: these are instances, but the capability is the reasoning. The second, and harder, is medicinal-chemical judgement — the ability to read a clinical need and recognise which molecules are worth developing at all. The first is learned by doing it under guidance; the second has to be earned through sustained work on one problem, and I am willing to give it that time.

Industry also changed how I read patents. As a student I found them verbose and difficult to interpret; three years in process development taught me that they are the most concentrated record of what a company genuinely believes is worth protecting. I now treat the patent landscape as part of the design problem — a compound that cannot be protected has limited value however potent it is. That is why I want to work where results are expected to survive contact with intellectual property and with the clinic. My longer-term aim is to work on molecules that reach patients; a doctoral project is where the reasoning behind such molecules is learned, not where they are finished.